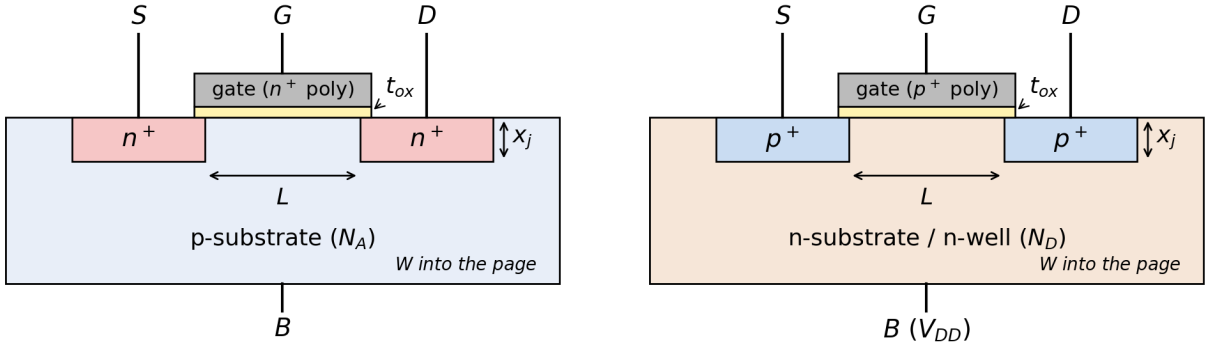
Exercise 03 – Solution Summary

Task 3.1

Idea: Same V_{T0} formula as Task 2.3, now for NMOS **and** PMOS. Gate doping decides Φ_{GC}, implantation shifts V_{T0} to the target.

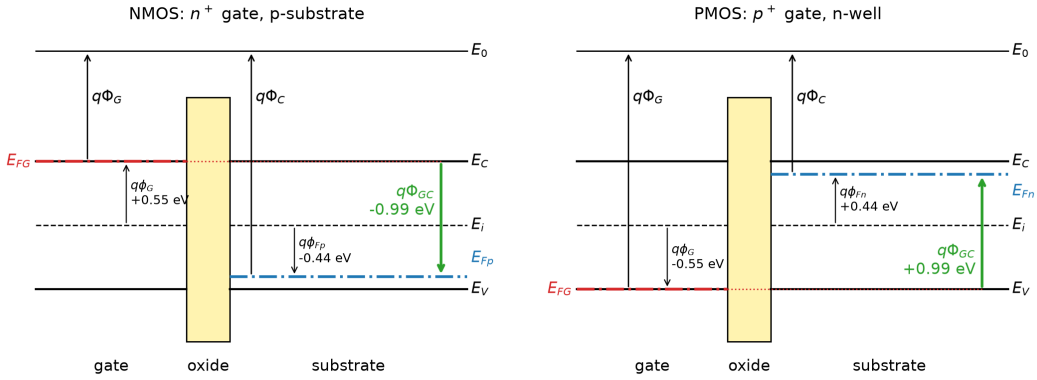
Structure of NMOS and PMOS:



Constants / given (all lengths in cm, values from Task 2.3):

- $kT/q = 0.026\,\text{V}, \quad q = 1.6 \cdot 10^{-19}\,\text{As}, \quad n_i = 1.45 \cdot 10^{10}\,\text{cm}^{-3}$
- $E_g = 1.1\,\text{eV} \rightarrow E_i$ in the middle, $0.55\,\text{eV}$ to each band edge
- $t_{ox} = 22\,\text{Å} = 22 \cdot 10^{-8}\,\text{cm} \rightarrow C_{ox} = \dfrac{\epsilon_{ox}}{t_{ox}} \approx 1.6 \cdot 10^{-6}\,\text{F/cm}^2$
- Substrate / well: $N_A = N_D = 3 \cdot 10^{17}\,\text{cm}^{-3}$, gate: $3 \cdot 10^{20}\,\text{cm}^{-3}$, $N_{SS} = 6 \cdot 10^{11}\,\text{cm}^{-2}$

1. **Task (a): V_{T0} of NMOS and PMOS (no implant, $V_{SB} = 0$)**
   - Equation (2.11):
   $$V_{T0} = \Phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$
   - Fermi potential, same doping → same magnitude:
   $$\phi_{Fp} = \frac{kT}{q} \ln \frac{n_i}{N_A} \approx -0.44\,\text{V}, \qquad \phi_{Fn} = \frac{kT}{q} \ln \frac{N_D}{n_i} \approx +0.44\,\text{V}$$
   - Work function difference $\Phi_{GC} = \phi_F - \phi_G$:
     - n^+ gate: $E_{FG} \approx E_C \rightarrow \phi_G = +0.55\,\text{V}$
     - p^+ gate: $E_{FG} \approx E_V \rightarrow \phi_G = -0.55\,\text{V}$



$$q\Phi_{GC} = E_{F,\text{sub}} - E_{FG}$$

NMOS: $\Phi_{GC} = \phi_{Fp} - \phi_G = -0.44\,\text{V} - 0.55\,\text{V} = -0.99\,\text{V}$

PMOS: $\Phi_{GC} = \phi_{Fn} - \phi_G = 0.44\,\text{V} + 0.55\,\text{V} = +0.99\,\text{V}$

- Depletion charge, magnitude from Task 2.3:

$$\left|\frac{Q_{B0}}{C_{ox}}\right| = \frac{\sqrt{2qN\epsilon_{si}\,|2\phi_F|}}{C_{ox}} \approx 0.188\,\text{V}$$

  - NMOS: ionized acceptors → $Q_{B0} < 0$
  - PMOS: ionized donors → $Q_{B0} > 0$

- Oxide charge, always positive:

$$\frac{Q_{ox}}{C_{ox}} = \frac{qN_{SS}}{C_{ox}} = \frac{1.6 \cdot 10^{-19}\,\text{As} \cdot 6 \cdot 10^{11}\,\text{cm}^{-2}}{1.6 \cdot 10^{-6}\,\text{F/cm}^2} = 0.06\,\text{V}$$

- **NMOS** (p-substrate, n^+ gate):

$$V_{T0n} = -0.99\,\text{V} - (-0.88\,\text{V}) - (-0.188\,\text{V}) - 0.06\,\text{V} \approx \boxed{0.02\,\text{V}}$$

- **PMOS** (n-well, p^+ gate):

$$V_{T0p} = 0.99\,\text{V} - 0.88\,\text{V} - 0.188\,\text{V} - 0.06\,\text{V} \approx \boxed{-0.14\,\text{V}}$$

Table 2.1
Signs in threshold voltage equation

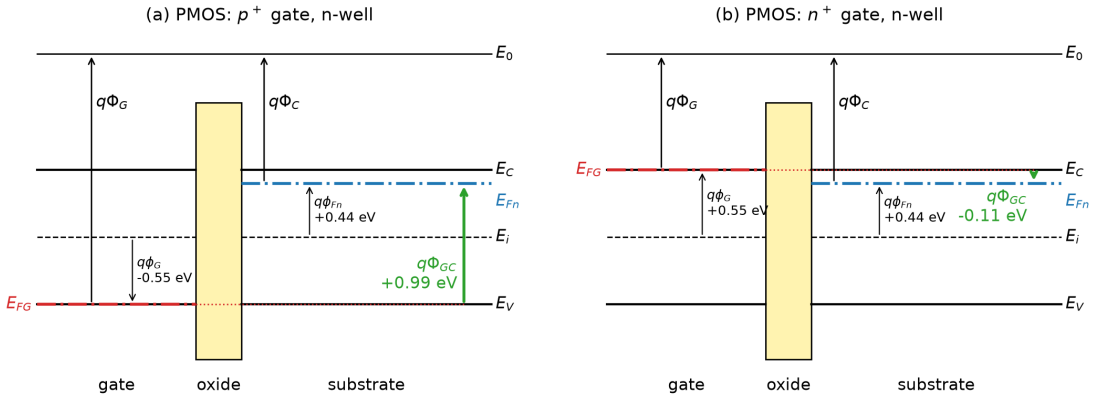
Parameter	NMOS	PMOS
Substrate	*p*-type	*n*-type
V_{T0}	+	−
ϕ_{GC}:		
n^+ polysilicon gate	−	−
p^+ polysilicon gate	+	+
Φ_F	−	+
Q_{B0}, Q_B	−	+
Q_{ox}	+	+
γ	+	−
X_d, C_{ox}	+	+
V_{SB}	+	−

Note: P50

2. **Task (b): PMOS with n^+ gate (donors instead of acceptors)**

- Donors → electrons, acceptors → holes
- n^+ gate → $\phi_G = +0.55\,\text{V}$:

$$\Phi_{GC} = \phi_{Fn} - \phi_G = 0.44\,\text{V} - 0.55\,\text{V} = -0.11\,\text{V}$$



E_{FG} moves from E_V (p^+) to E_C (n^+): Φ_{GC} drops from $+0.99\,\text{V}$ to $-0.11\,\text{V}$, i.e. by $E_g/q = 1.1\,\text{V}$

- Rest unchanged:

$$V_{T0p} = -0.11\,\text{V} - 0.88\,\text{V} - 0.188\,\text{V} - 0.06\,\text{V} \approx \boxed{-1.24\,\text{V}}$$

3. **Task (c): Implant dose N_I for $V_{T0n} = 0.4\,\text{V}$, $V_{T0p} = -0.4\,\text{V}$**

- Implanted charge qN_I at the surface, like a capacitor $Q = C_{ox}\,\Delta V$:

$$N_I = \frac{C_{ox}}{q}\,\Delta V_T, \qquad \frac{C_{ox}}{q} = \frac{1.6 \cdot 10^{-6}\,\text{F/cm}^2}{1.6 \cdot 10^{-19}\,\text{As}} = 10^{13}\,\text{V}^{-1}\text{cm}^{-2}$$

- Sign convention: $N_I > 0$ → acceptors (V_T up), $N_I < 0$ → donors (V_T down)
- NMOS (n^+ gate):

$$N_I = 10^{13}\,\text{V}^{-1}\text{cm}^{-2} \cdot (0.4\,\text{V} - 0.02\,\text{V}) = \boxed{3.8 \cdot 10^{12}\,\text{cm}^{-2}} \quad \text{(acceptors)}$$

- PMOS (p^+ gate):

$$N_I = 10^{13}\,\text{V}^{-1}\text{cm}^{-2} \cdot (-0.4\,\text{V} - (-0.14\,\text{V})) = \boxed{-2.6 \cdot 10^{12}\,\text{cm}^{-2}} \quad \text{(donors)}$$

- PMOS (n^+ gate, from (b)):

$$N_I = 10^{13}\,\text{V}^{-1}\text{cm}^{-2} \cdot (-0.4\,\text{V} - (-1.24\,\text{V})) = \boxed{8.4 \cdot 10^{12}\,\text{cm}^{-2}} \quad \text{(acceptors)}$$

4. **Task (d): Why n^+ gate for NMOS and p^+ gate for PMOS?**

- $0.13\,\mu$m node: $V_{DD} = 1.2\,\text{V}$ → need $|V_{T0}| < V_{DD}$

- Matching gate: $\Phi_{GC} \approx -2\phi_F \rightarrow |V_{T0}|$ small ($0.02\,\text{V}$, $-0.14\,\text{V}$) → small implant
- Wrong gate: $|V_{T0}| \approx 1.2\,\text{V}$ (Task 2.3: $1.18\,\text{V}$, (b): $-1.24\,\text{V}$) → $\geq V_{DD}$ → transistor never on

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5. **Task (e): Effective mobility μ_e of the PMOS**

   - Equation (2.21), P59, mobility drops with vertical field:

$$\mu_e = \frac{\mu_0}{1 + \left(\frac{V_{GS} - V_T}{\theta \cdot t_{ox}}\right)^{\eta}}$$

   - Given: $\mu_0 = 130\,\text{cm}^2\text{V}^{-1}\text{s}^{-1}, \quad \theta = 4 \cdot 10^6\,\text{Vcm}^{-1}, \quad \eta = 1.85$
     - μ_0: low-field mobility (holes), upper limit for $E_{vert} \rightarrow 0$
     - θ: critical vertical field, $E_{vert} = \theta \rightarrow \mu_e = \mu_0/2$
     - η: fitting exponent, how steep μ_e drops with E_{vert}
   - PMOS: $V_{SG} - |V_{TP}| = 1.2\,\text{V} - 0.4\,\text{V} = 0.8\,\text{V}$
     - $V_{SG} = V_{DD} = 1.2\,\text{V}$ (0.13 µm, fully on → max. vertical field), $|V_{TP}| = 0.4\,\text{V}$ from (c)
   - Insert:

$$\mu_e = \frac{130\,\text{cm}^2\text{V}^{-1}\text{s}^{-1}}{1 + \left(\frac{0.8\,\text{V}}{4 \cdot 10^6\,\text{Vcm}^{-1} \cdot 22 \cdot 10^{-8}\,\text{cm}}\right)^{1.85}} = \frac{130\,\text{cm}^2\text{V}^{-1}\text{s}^{-1}}{1 + 0.91^{1.85}} \approx \boxed{71\,\text{cm}^2\text{V}^{-1}\text{s}^{-1}}$$

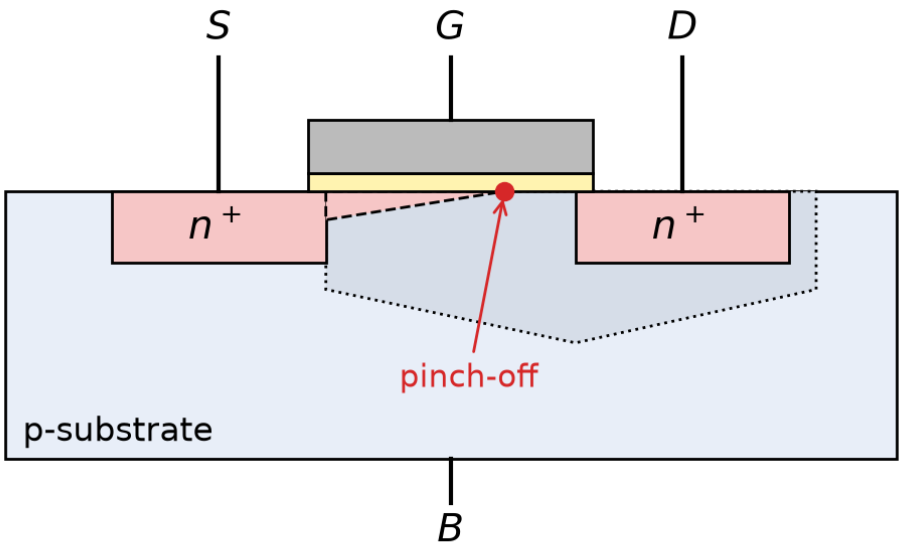
   Note:

   - $22 \cdot 10^{-8}\,\text{cm}$ in the denominator is $t_{ox} = 22\,\text{Å}$
   - $E_{vert} = \frac{V_{GS} - V_T}{t_{ox}} = \frac{0.8\,\text{V}}{22 \cdot 10^{-8}\,\text{cm}} \approx 3.6 \cdot 10^6\,\text{V/cm} \approx \theta$

Task 3.2

Idea: Short channel (130nm, 180nm) → velocity saturation: carriers reach v_{sat} before pinch-off → $V_{Dsat} < V_{GS} - V_T$.

Background: pinch-off (long channel)



saturation: $V_{GS} > V_T$, $V_{DS} > V_{GS} - V_T$

- Two fields in the channel:
  - **Vertical** $E_{vert} \approx (V_{GS} - V_{ch} - V_T)/t_{ox}$ (gate → channel): decides how many electrons, $Q_n(x) = -C_{ox}(V_{GS} - V_{ch}(x) - V_T)$ → Task 3.1 (e)
  - **Lateral** $E_{lat} \approx dV_{ch}/dx$ (from V_{DS}, along S → D): moves the electrons → current → velocity saturation E_c
- V_{ch} rises from 0 (S) to V_{DS} (D) → $V_{GS} - V_{ch}$ shrinks towards D → weaker E_{vert} → channel gets thinner
- $V_{DS} = V_{GS} - V_T$: at the drain end $V_{GS} - V_{ch} = V_T \to Q_n \to 0$ → **pinch-off** at D, $V_{Dsat} = V_{GS} - V_T$
- $V_{DS} > V_{Dsat}$: P moves towards S by ΔL; the extra $V_{DS} - (V_{GS} - V_T)$ drops over the short ΔL → very high E_{lat} sweeps electrons into D → $I_D \approx$ const. (saturation)
- Short channel: $E_{lat} \approx V_{DS}/L$ reaches E_c before pinch-off → $v = v_{sat}$ → saturation earlier, $V_{Dsat} < V_{GS} - V_T$

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1. **Task (a):** V_{Dsat} **in** $0.18\,\mu$**m (**$V_{DD} = 1.8$ **V,** $L = 200$ **nm,** $|V_T| = 0.5$ **V)**

* Critical field: $E_{cn} = 6 \cdot 10^4$ V/cm (electrons), $E_{cp} = 24 \cdot 10^4$ V/cm (holes), equation (2.22), P60 * Equation (2.28), P62:

$$V_{Dsat} = \frac{(V_{GS} - V_T)\, E_c L}{(V_{GS} - V_T) + E_c L}$$

- $E_c L$:

  $E_{cn}L = 6{\cdot}10^4\,\text{V/cm}{\cdot}200{\cdot}10^{-7}\,\text{cm} = 1.2\,\text{V}, \qquad E_{cp}L = 24{\cdot}10^4\,\text{V/cm}{\cdot}200{\cdot}10^{-7}\,\text{cm} = 4.8\,\text{V}$

- NMOS:

$$V_{Dsat} = \frac{(1.8\,\text{V} - 0.5\,\text{V}) \cdot 1.2\,\text{V}}{1.8\,\text{V} - 0.5\,\text{V} + 1.2\,\text{V}} \approx \boxed{0.6\,\text{V}}$$

- PMOS:

$$V_{Dsat} = \frac{(1.8\,\text{V} - 0.5\,\text{V}) \cdot 4.8\,\text{V}}{1.8\,\text{V} - 0.5\,\text{V} + 4.8\,\text{V}} \approx \boxed{1.0\,\text{V}}$$

2. **Task (b): V_{Dsat} in $0.13\,\mu$m ($E_{cn}L_n = 0.6$ V, $E_{cp}L_p = 2.4$ V, $V_{DD} = 1.2$ V, $|V_T| = 0.4$ V)**

   - NMOS:
   $$V_{Dsat} = \frac{(1.2\,\text{V} - 0.4\,\text{V}) \cdot 0.6\,\text{V}}{1.2\,\text{V} - 0.4\,\text{V} + 0.6\,\text{V}} \approx \boxed{0.34\,\text{V}}$$
   - PMOS:
   $$V_{Dsat} = \frac{(1.2\,\text{V} - 0.4\,\text{V}) \cdot 2.4\,\text{V}}{1.2\,\text{V} - 0.4\,\text{V} + 2.4\,\text{V}} = \boxed{0.6\,\text{V}}$$

   Note: Long channel: $V_{Dsat} = V_{GS} - V_T$

3. **Task (c): Ratio of saturation currents in $0.13\,\mu$m**

   - Equation (2.29), P63:
   $$I_{Dsat} = W v_{sat} C_{ox} \frac{(V_{GS} - V_T)^2}{(V_{GS} - V_T) + E_c L}$$
   - $v_{sat}C_{ox} = 8 \cdot 10^6$ cm/s $\cdot\, 1.6 \cdot 10^{-6}$ F/cm^2 $= 12.8\,\text{A}\,(\text{Vcm})^{-1}$
   - NMOS per width:
   $$\frac{I_{DsatN}}{W} = 12.8\,\text{A}\,(\text{Vcm})^{-1} \cdot \frac{(0.8\,\text{V})^2}{0.8\,\text{V} + 0.6\,\text{V}} \approx 5.85\,\text{A/cm} = 585\,\mu\text{A}/\mu\text{m}$$
   - PMOS per width:
   $$\frac{I_{DsatP}}{W} = 12.8\,\text{A}\,(\text{Vcm})^{-1} \cdot \frac{(0.8\,\text{V})^2}{0.8\,\text{V} + 2.4\,\text{V}} \approx 2.56\,\text{A/cm} = 256\,\mu\text{A}/\mu\text{m}$$
   - Same W, same $|V_{GS} - V_T| \rightarrow$ only E_cL differs:
   $$\frac{I_{DsatN}}{I_{DsatP}} = \frac{(V_{SG} - |V_{TP}|) + E_{cp}L_p}{(V_{GS} - V_{TN}) + E_{cn}L_n} = \frac{0.8\,\text{V} + 2.4\,\text{V}}{0.8\,\text{V} + 0.6\,\text{V}} \approx \boxed{2.3}$$

   Note:

   - $1\,\text{A/cm} = 100\,\mu\text{A}/\mu\text{m}$
   - NMOS ≈ 2.3× stronger → balance by $W_P \approx 2.3\,W_N$ ($I_{Dsat} \propto W$)

Task 3.3

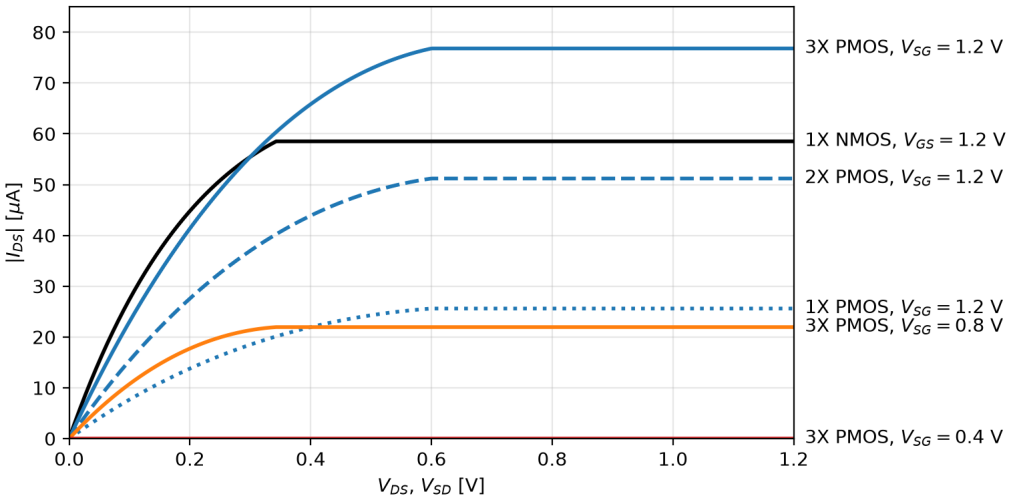
Idea: All currents relative to a 1X PMOS at $V_{SG} = 1.2$ V: scale with W ($I \propto W$) and with V_{GS} via equation (2.29).

Given: $0.13\,\mu$m, $V_{DD} = 1.2$ V, $1X$: $W = L = 100$ nm, $|V_T| = 0.4$ V.

- Equation (2.29), P63:
$$I_{Dsat} = W v_{sat} C_{ox} \frac{(V_{GS} - V_T)^2}{(V_{GS} - V_T) + E_c L}$$

1. **Task (a):** I_{DS} **over** V_{DS}
   - From Task 3.2 (c): 1X NMOS $= 2.3\cdot$ 1X PMOS
   - $2X$, $3X$ PMOS $\rightarrow 2\times, 3\times$ current
   - $V_{SG} = 0.4\,\text{V} = |V_{TP}| \rightarrow$ cutoff, $I_{DS} \approx 0$ (only subthreshold current)
   - Relative saturation currents (1X PMOS, $V_{SG} = 1.2\,\text{V}$ = 1):

Device	I_{Dsat} (rel.)	I_{Dsat} ($W = 100$ nm)	V_{Dsat}
3X PMOS, $V_{SG} = 1.2\,\text{V}$	3	$77\,\mu$A	$0.6\,\text{V}$
1X NMOS, $V_{GS} = 1.2\,\text{V}$	2.3	$59\,\mu$A	$0.34\,\text{V}$
2X PMOS, $V_{SG} = 1.2\,\text{V}$	2	$51\,\mu$A	$0.6\,\text{V}$
1X PMOS, $V_{SG} = 1.2\,\text{V}$	1	$26\,\mu$A	$0.6\,\text{V}$
3X PMOS, $V_{SG} = 0.8\,\text{V}$	$3 \cdot 0.29 \approx 0.86$	$22\,\mu$A	$0.34\,\text{V}$
3X PMOS, $V_{SG} = 0.4\,\text{V}$	0	0	–



Note:

- Plot: velocity-saturation model, $\lambda = 0 \rightarrow$ flat in saturation; real curves rise slightly (channel length modulation)
- Official sketch (Fig. 3.2) is copied from the textbook, P64, not accurate

2. **Task (b): NMOS** I_{DS} **over** V_{GS}**,** $V_{DS} = 1.2\,\text{V}$
   - Operating regions (long-channel, λ neglected):
     - $V_{GS} \leq V_{TN}$: cutoff $\rightarrow I_{DS} = 0$

  - $V_{DS} < V_{GS} - V_{TN}$: linear, equation (2.17b), P54:

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left(V_{GS} - V_{TN} - \frac{V_{DS}}{2} \right) V_{DS}$$

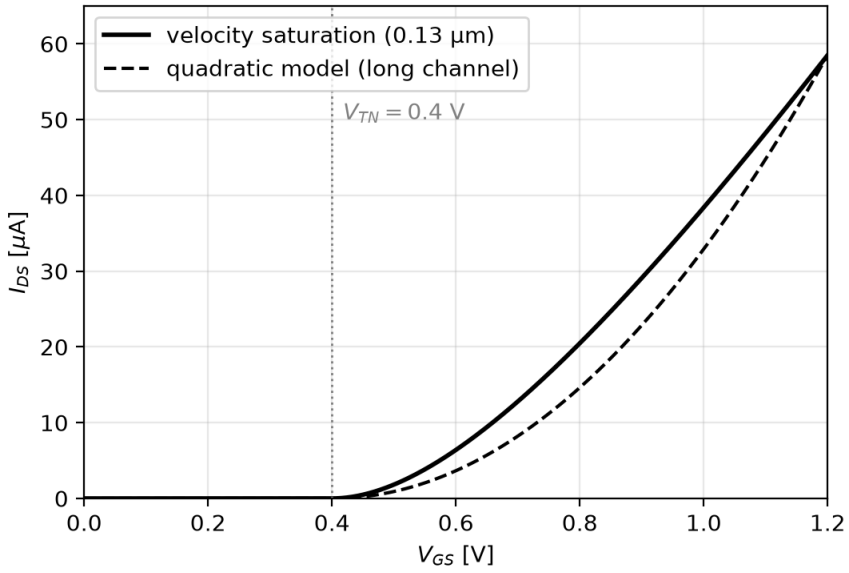
  - $V_{DS} \geq V_{GS} - V_{TN}$: saturation, equation (2.19), P54:

$$I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TN})^2$$

- $V_{DS} = 1.2\,\text{V} > V_{Dsat}$ for all $V_{GS} \leq 1.2\,\text{V}$ → always saturation
- Velocity saturation, equation (2.29), P63, $E_{cn} L_n = 0.6\,\text{V}$:

$$I_{DS} \propto \frac{(V_{GS} - V_{TN})^2}{(V_{GS} - V_{TN}) + 0.6\,\text{V}}$$

  - $V_{GS} - V_{TN} \ll E_{cn} L_n \rightarrow \propto (V_{GS} - V_{TN})^2$, quadratic
  - $V_{GS} - V_{TN} \gtrsim E_{cn} L_n \rightarrow \propto (V_{GS} - V_{TN})$, nearly linear



- **Quadratic model no longer valid** in $0.13\,\mu$m: I_{DS} almost linear in V_{GS} above V_{TN}

Task 3.4

Idea: Alpha power law $I_{DS} = K_S \frac{W}{L} (V_{GS} - V_T)^\alpha$. Ratio of two saturation currents.

Given: $W/L = 1$, $V_T = 0.5\,\text{V}$, Fig. 3.1.

1. **Read saturation currents from Fig. 3.1**
   - $V_{GS} = 1.35\,\text{V}$: $I_{DS} \approx 130\,\mu\text{A}$
   - $V_{GS} = 1.8\,\text{V}$: $I_{DS} \approx 220\,\mu\text{A}$

2. **Exponent α**

$$\frac{220\,\mu\text{A}}{130\,\mu\text{A}} = \frac{K_S(1.8\,\text{V} - 0.5\,\text{V})^\alpha}{K_S(1.35\,\text{V} - 0.5\,\text{V})^\alpha}$$
$$1.69 = 1.53^\alpha$$
$$\alpha = \frac{\log 1.69}{\log 1.53} \approx \boxed{1.25}$$

3. **Coefficient K_S**

$$130\,\mu\text{A} = K_S(0.85\,\text{V})^{1.25} \;\Rightarrow\; K_S \approx 159\,\mu\text{A/V}^{1.25}$$
$$220\,\mu\text{A} = K_S(1.3\,\text{V})^{1.25} \;\Rightarrow\; K_S \approx 158\,\mu\text{A/V}^{1.25}$$
$$\Rightarrow\; \boxed{K_S \approx 160\,\mu\text{A/V}^{1.25}}$$

4. **Check with the third curve ($V_{GS} = 0.9\,\text{V}$)**

$$I_{DS} = 160\,\frac{\mu\text{A}}{\text{V}^{1.25}} \cdot (0.9\,\text{V} - 0.5\,\text{V})^{1.25} \approx 50\,\mu\text{A}$$

- Matches Fig. 3.1 ($\approx 50\,\mu$A)